جامعة حفر الباطن
University of Hafr Al Batin

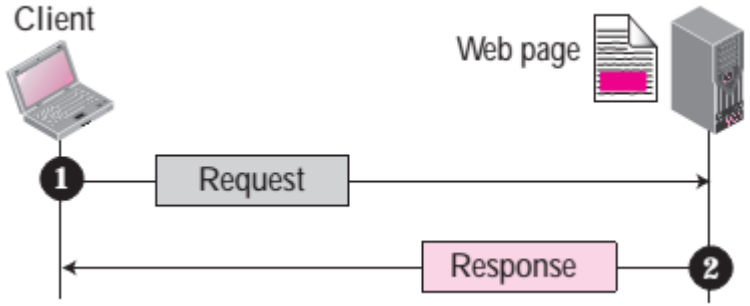
Lecture 14 - World Wide Web and HTTP

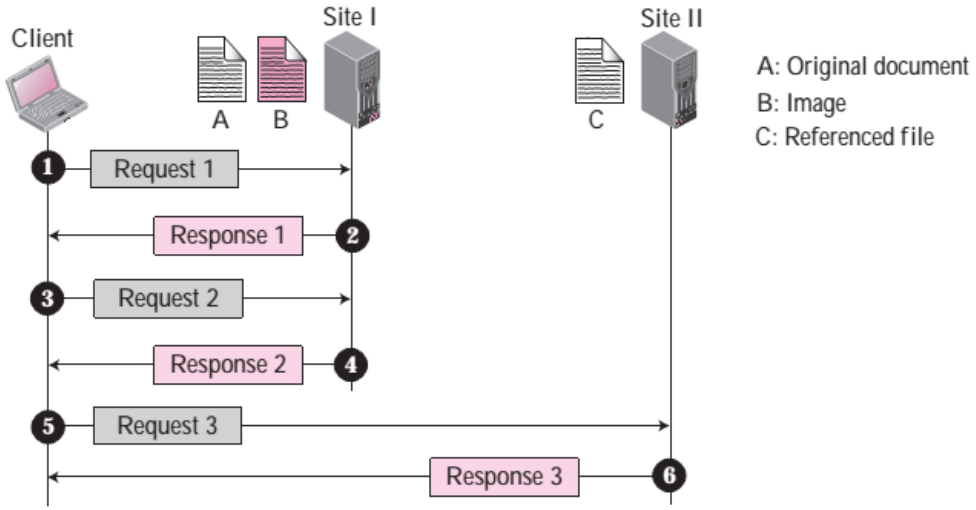
Objectives

- WWW Architecture.
- Web Client & Web Server.
- Web Documents.
- HTTP Overview.
- HTTP Transaction.
- HTTP Persistence.
- Cookies
- HTTP Security

Architecture

- The WWW today is a distributed client-server service, in which a client using a browser can access a service using a server.
  - The service provided is distributed over many locations called sites.
  - Each site holds one or more documents, referred to as Web pages.
  - Each Web page can contain some links to other Web pages in the same or other sites.



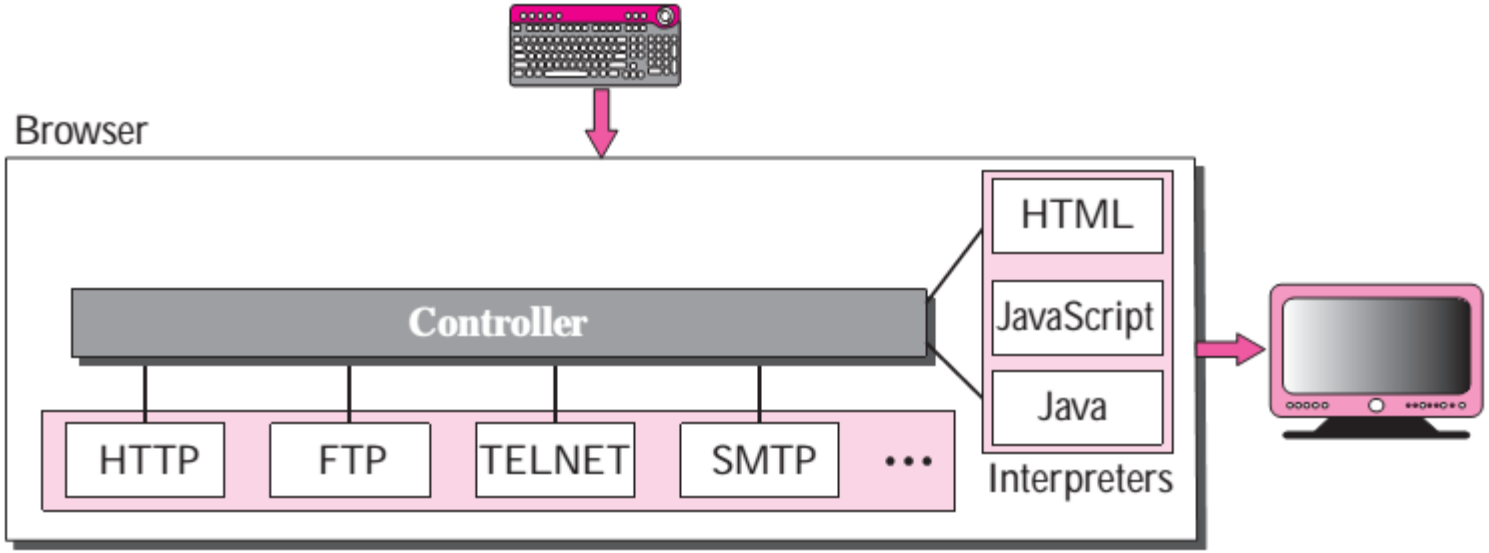


Hypertext and Hypermedia

- **Hypertext:** it means creating documents that refer to other documents.
  - In a hypertext document, a part of text can be defined as a link to another document.
  - When a hypertext is viewed with a browser, the link can be clicked to retrieve the other document.
- **Hypermedia:** is a term applied to document that contains links to other textual document or documents containing graphics, video, or audio.

Web Client (Browser)

- Each browser usually consists of three parts:
  1. Controller
  2. Client protocol
  3. Interpreters.



Web Server

- The Web page is stored at the server.
- Each time a client request arrives, the corresponding document is sent to the client.
- To improve **Server Efficiency**:
  1. A server stores requested files in a cache in memory.
     - memory is faster to access than disk.
  2. A server can use multithreading or multiprocessing.
     - In this case, a server can answer more than one request at a time.
- Some popular Web servers”
  - Apache.
  - Microsoft Internet Information Server.

Uniform Resource Locator (URL)

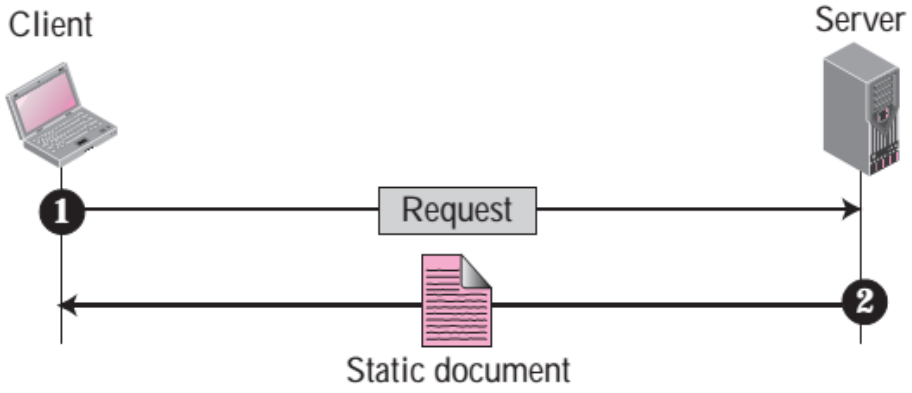
- The uniform resource locator (URL): is a standard locator for specifying any kind of information on the Internet.
- The URL defines four things:
  1. Protocol: the client-server application program used to retrieve the document.
  2. Host: the domain name of the computer on which the information is located.
  3. Port: port number of the server
  4. Path: the pathname of the file where the information is located

Web Documents

The documents in the WWW can be grouped into three broad categories:

1. Static documents: are fixed-content documents that are created and stored in a server.

- Static documents are prepared using one of the several languages:
  - Hypertext Markup Language (HTML).
  - Extensible Markup Language (XML).
  - Extensible Style Language (XSL).
  - Extended Hypertext Markup Language (XHTML).

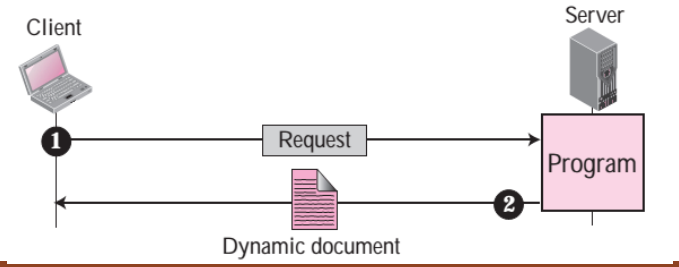


Web Documents

The documents in the WWW can be grouped into three broad categories:

2. Dynamic document: it is created by a Web server whenever a browser requests the document.

- When a request arrives:
  - Server runs an application program or a script that creates the dynamic document.
  - Server returns the output of the program or script as a response to the browser.
  - *Example*: the retrieval of the time and date from a server.
  - Dynamic documents are sometimes referred to as *server-site dynamic documents*.



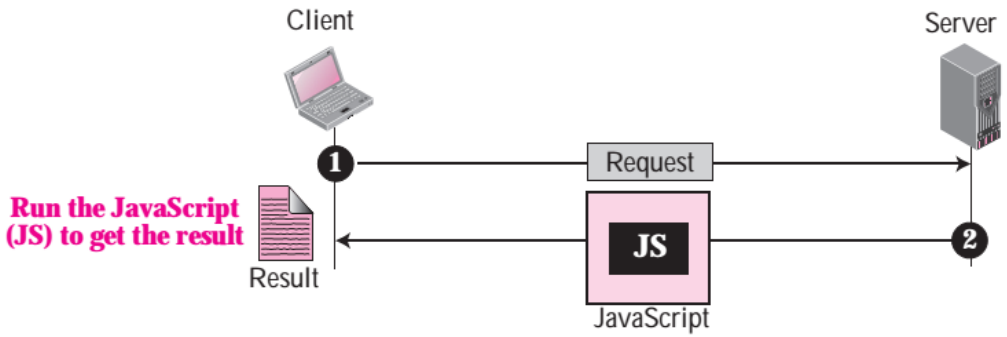
Web Documents

The documents in the WWW can be grouped into three broad categories:

2. Active document: a program or a script that runs at the client site.

When a browser request an active document:

- the server sends a copy of the document or a script.
- The document is then run at the client (browser) site.
- Active documents are sometimes referred to as *client-site dynamic documents*.

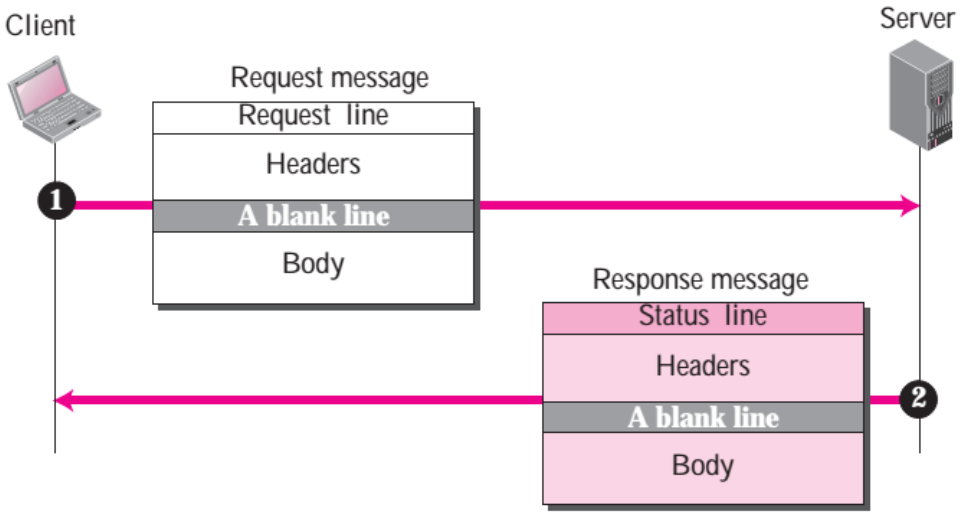


HTTP

- **Hypertext Transfer Protocol (HTTP):** is a protocol used mainly to access data on the World Wide Web.
- HTTP functions like a combination of FTP and SMTP.
- It is similar to FTP because it transfers files and uses TCP.
  - **HTTP uses the services of TCP on well-known port 80.**
- However, it is much simpler than FTP because it uses only one TCP connection.
  - There is no separate control connection; only data are transferred between the client and the server.

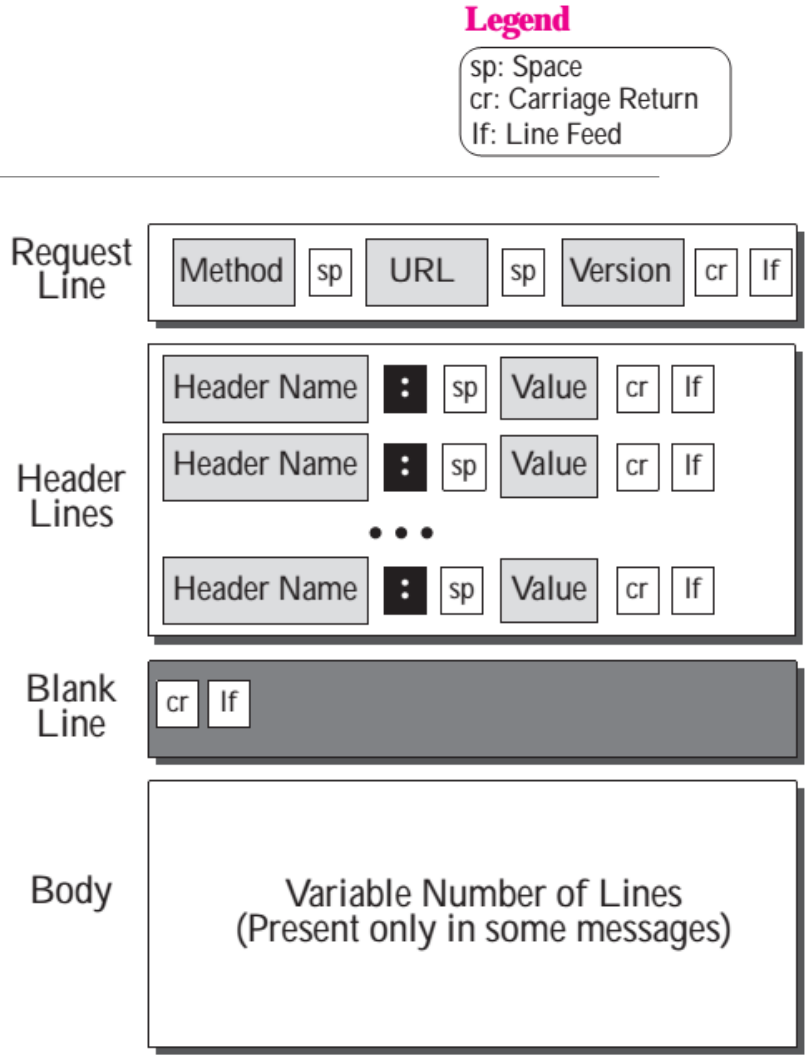
HTTP Transaction

- The client initializes the transaction by sending a request.
- The server replies by sending a response
- **HTTP is a stateless protocol.**
  - which means that the server does not keep information about the client.



HTTP Transaction (Request)

- A request message consists of three parts:
  1. Request line.
     - Method: the request type.
     - URL: the address of the web page.
     - Version: the http protocol version (current version 1.1)
  2. Header Lines.
     - Additional information from the client to the server.
  3. Body (optional).
     - Usually, it contains the comment to be sent.



HTTP Transaction (Request)

- HTTP 1.1 methods:

Method	*Action*
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
POST	Sends some information from the client to the server
PUT	Sends a document from the server to the client
TRACE	Echoes the incoming request
CONNECT	Reserved
DELETE	Remove the Web page
OPTIONs	Enquires about available options

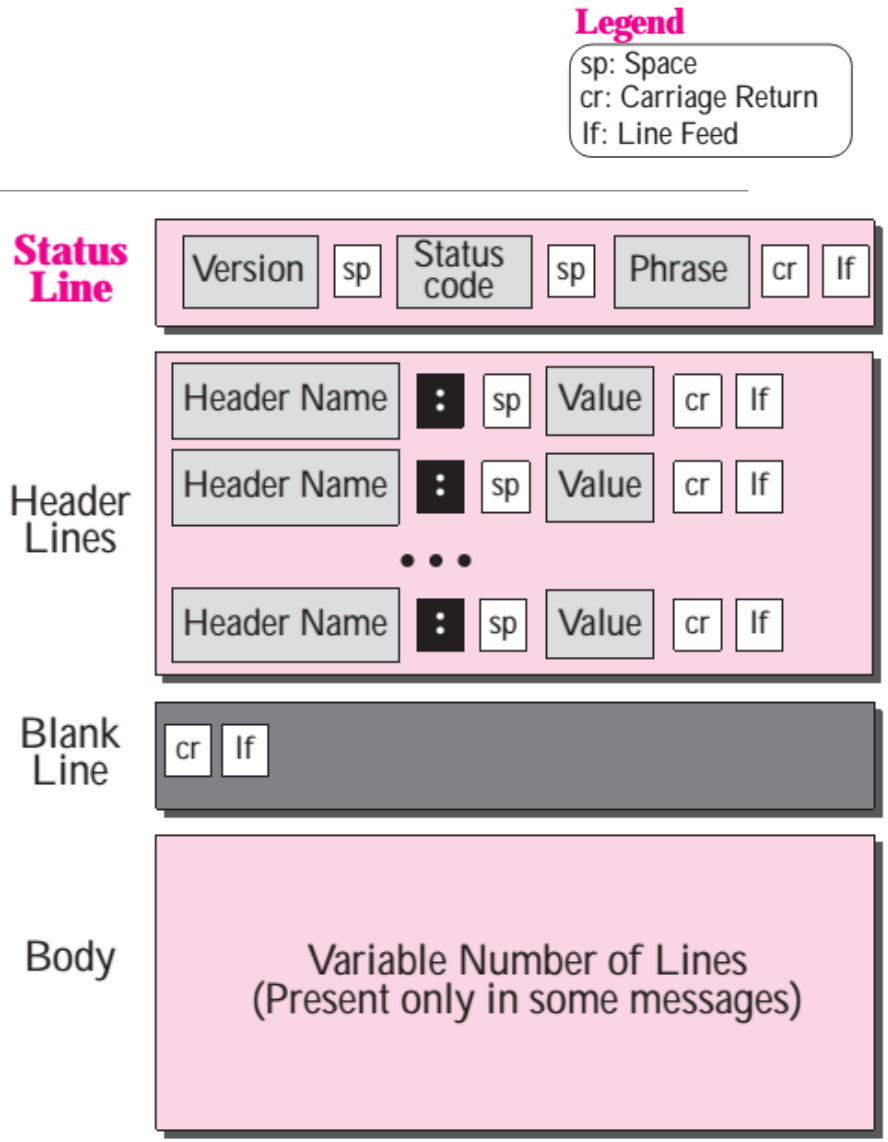
HTTP Transaction (Request)

- Request Header Names:

Header	*Description*
User-agent	Identifies the client program
Accept	Shows the media format the client can accept
Accept-charset	Shows the character set the client can handle
Accept-encoding	Shows the encoding scheme the client can handle
Accept-language	Shows the language the client can accept
Authorization	Shows what permissions the client has
Host	Shows the host and port number of the client
Date	Shows the current date
Upgrade	Specifies the preferred communication protocol
Cookie	Returns the cookie to the server
If-Modified-Since	Returns the cookie to the server

HTTP Transaction (Response)

- A response message consists of:
  1. A status line
     - Version: version of HTTP protocol.
     - Status code: the status of the request (3 digits).
     - Status phrase: the status code in text form.
  2. Header Lines.
     - Additional information from the server to the client.
  3. A blank line
  4. Body
     - Contains the document to be sent from the server to the client.
     - The body is present unless the response is an error message.



HTTP Transaction (Response)

- Response **Status Codes** Types:
  1. Informational (100s)
  2. Success (200s)
  3. Redirection (300s)
  4. Client Error (400s)
  5. Server Error (500s)

Status Code	*Status Phrase*	*Description*
	Informational	
100	Continue	The initial part of the request received, continue.
101	Switching	The server is complying to switch protocols.
	Success	
200	OK	The request is successful.
201	Created	A new URL is created.
202	Accepted	The request is accepted, but it is not immediately acted upon.
204	No content	There is no content in the body.
	Redirection	
301	Moved permanently	The requested URL is no longer used by the server.
302	Moved temporarily	The requested URL has moved temporarily.
304	Not modified	The document has not modified.
	Client Error	
400	Bad request	There is a syntax error in the request.
401	Unauthorized	The request lacks proper authorization.
403	Forbidden	Service is denied.
404	Not found	The document is not found.
405	Method not allowed	The method is not supported in this URL.
406	Not acceptable	The format requested is not acceptable.
	Server Error	
500	Internal server error	There is an error, such as a crash, at the server site.
501	Not implemented	The action requested cannot be performed.
503	Service unavailable	The service is temporarily unavailable.

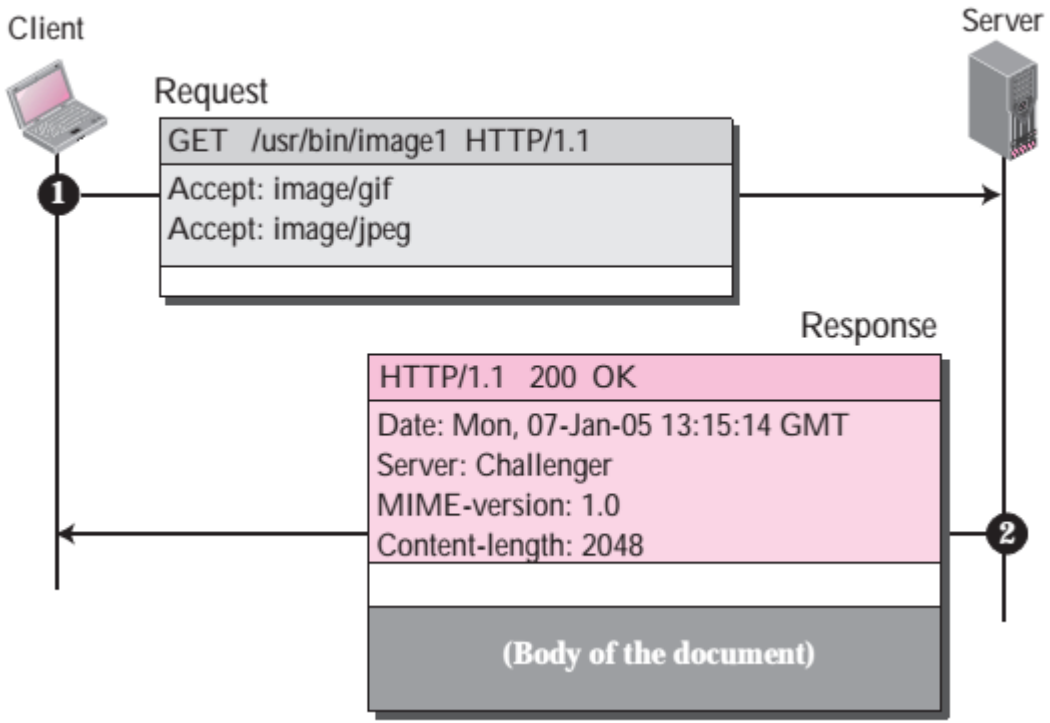
HTTP Transaction (Response)

- Response Header Names:

Header	*Description*
Date	Shows the current date
Upgrade	Specifies the preferred communication protocol
Server	Gives information about the server
Set-Cookie	The server asks the client to save a cookie
Content-Encoding	Specifies the encoding scheme
Content-Language	Specifies the language
Content-Length	Shows the length of the document
Content-Type	Specifies the media type
Location	To ask the client to send the request to another site
Accept-Ranges	The server will accept the requested byte-ranges
Last-modified	Gives the date and time of the last change

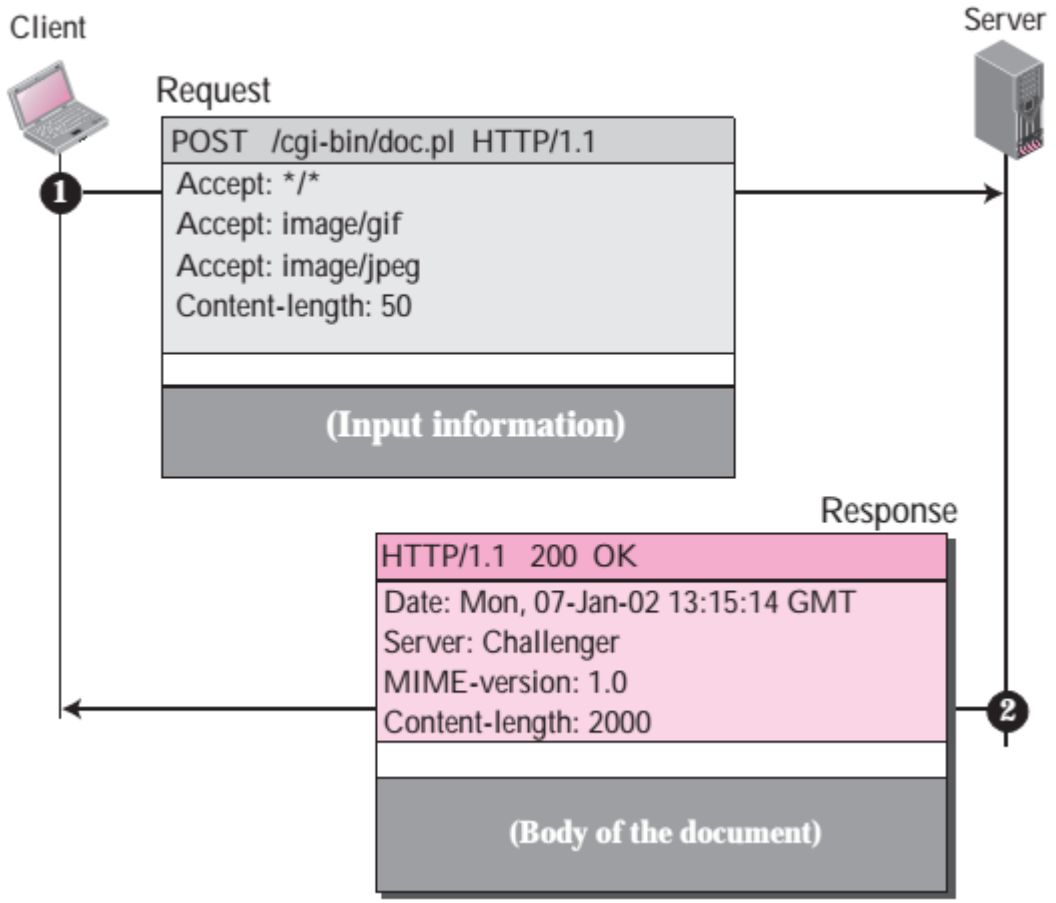
HTTP Transaction

- Example:
  - Retrieve a document.



HTTP Transaction

- Example:
  - Client sends data to the server.

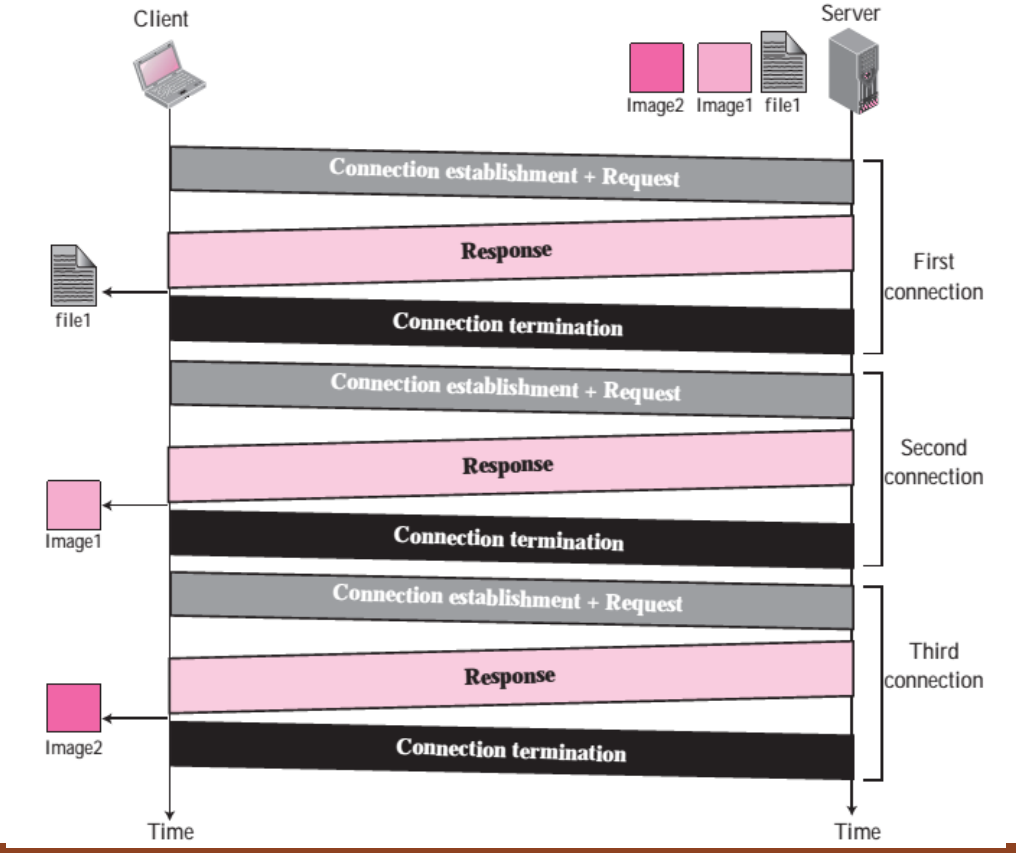


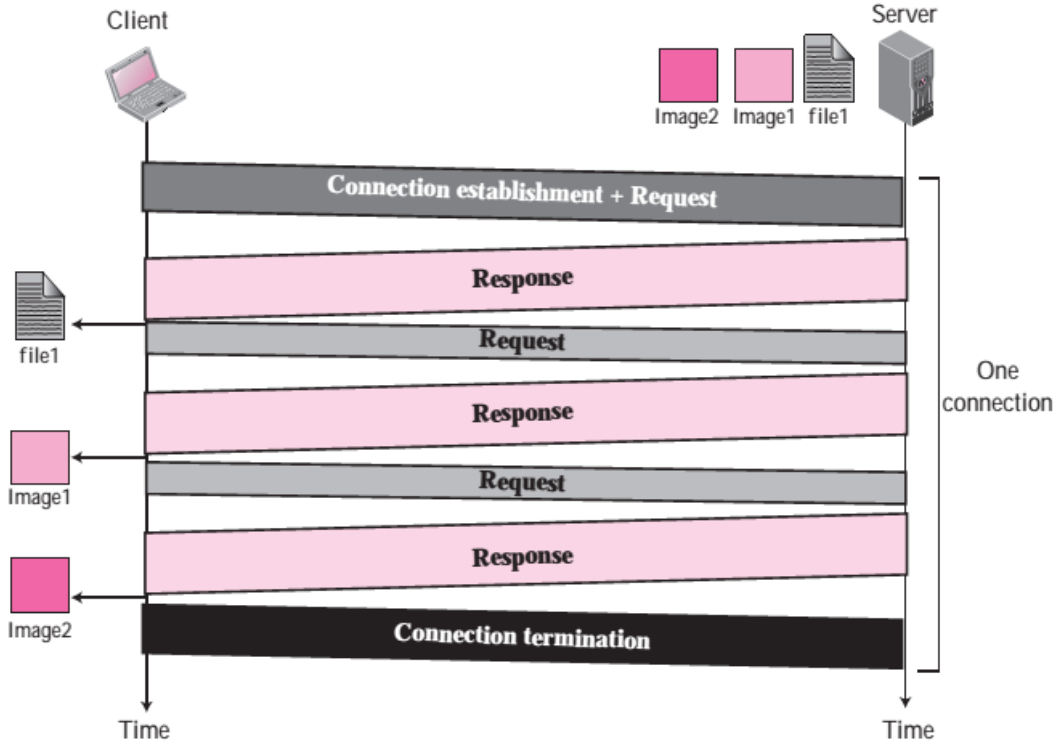
HTTP Persistence

- HTTP 1.1 uses persistent connection as default.
  - Prior to version 1.1, HTTP specified a nonpersistent connection.
- **Nonpersistent Connection:** The server closes the connection after fulfilling the request (after sending the response).
  - if a file contains link to N different pictures in different files (all located on the same server), the connection must be opened and closed N + 1 times.
- **Persistent Connection:** the server leaves the connection open for more requests after sending a response.
  - The server can close the connection at the request of a client or if a time-out has been reached.

HTTP Persistence

- Nonpersistent vs Persistent Connection:



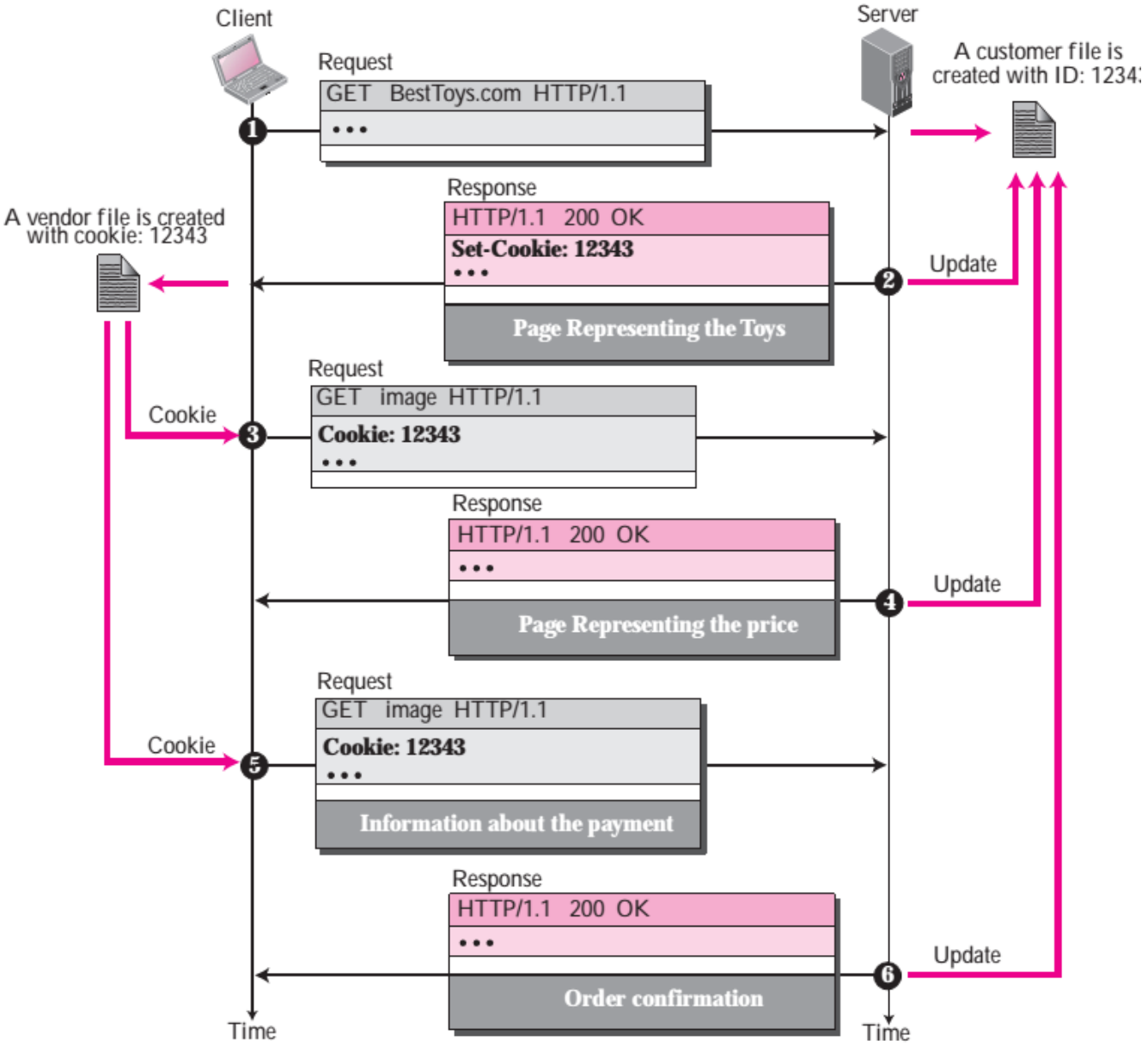


Cookies

- **Creating and storing Cookies:**
  1. When a server receives a request from a client, it stores information about the client in a file or a string.
  2. The server includes the cookie in the response that it sends to the client.
  3. The client browser stores the cookie in the cookie directory..
- **Using Cookies:**
  1. When a client sends a request to a server, the browser looks in the cookie directory to see if it can find a cookie sent by that server.
  2. If found, the cookie is included in the request.
  3. When the server receives the request, it knows that this is an old client, not a new one.
- Cookie content are never read by the browser or disclosed to the user.
  - It is a cookie *made* by the server and *eaten* by the server.

Cookies

- Example:
  - Buying from an electronic store.



HTTP Security

- The HTTP does not provide security.
- However, HTTP can be run over the Secure Socket Layer (SSL).
  - In this case, HTTP is referred to as HTTPS.
- **HTTPS** provides:
  1. Confidentiality.
  2. Client and server authentication.
  3. Data integrity.